4.7.9 Quality factor or Q–factor

The current in the series *RLC* circuit becomes maximum at resonance. Due to the increase in current, the voltage across *L* and *C* are also increased. This magnification of voltages at series resonance is termed as Q–factor.

It is defined as the ratio of voltage across *L* or *C* at resonance to the applied voltage.

$$\text{Q-factor} = \frac{\text{Voltage across L or C at resonance}}{\text{Applied voltage}}$$

At resonance, the circuit is purely resistive. Therefore, the applied voltage is equal to the voltage across *R*.

$$\text{Q-factor} = \frac{I_m X_L}{I_m R} = \frac{X_L}{R} \qquad (4.52)$$

$$= \frac{\omega_r L}{R}$$

$$= \frac{L}{R\sqrt{LC}} \qquad \text{since } \omega_r = \frac{1}{\sqrt{LC}}$$

$$\text{Q-factor} = \frac{1}{R}\sqrt{\frac{L}{C}} \qquad (4.53)$$

The physical meaning is that Q–factor indicates the number of times the voltage across *L* or *C* is greater than the applied voltage at resonance.

EXAMPLE 4.22

Find the impedance of a series *RLC* circuit if the inductive reactance, capacitive reactance and resistance are 184 Ω, 144 Ω and 30 Ω respectively. Also calculate the phase angle between voltage and current.

Solution

X_L = 184 Ω; X_C = 144 Ω

R = 30 Ω

(i) The impedance is

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{30^2 + (184 - 144)^2}$$

$$= \sqrt{900 + 1600}$$

$$Z = 50\ \Omega$$

(ii) Phase angle ϕ between voltage and current is

$$\tan\phi = \frac{X_L - X_C}{R}$$

$$= \frac{184 - 144}{30} = 1.33$$

$$\phi = 53.1^\circ$$

Since the phase angle is positive, voltage leads current by 53.1° for this inductive circuit.

EXAMPLE 4.23

A 500 μH inductor, $\frac{80}{\pi^2} pF$ capacitor and a 628 Ω resistor are connected to form a series *RLC* circuit. Calculate the resonant frequency and Q-factor of this circuit at resonance.

Solution

$$L = 500 \times 10^{-6}\text{H}; C = \frac{80}{\pi^2} \times 10^{-12}\text{F}; \quad R = 628\,\Omega$$

(i) Resonant frequency is

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{500 \times 10^{-6} \times \frac{80}{\pi^2} \times 10^{-12}}}$$

$$= \frac{1}{2\sqrt{40{,}000 \times 10^{-18}}}$$

$$= \frac{10{,}000 \times 10^3}{4} = 2500\ \text{KHz}$$

(ii) *Q*–factor

$$= \frac{\omega_r L}{R} = \frac{2 \times 3.14 \times 2500 \times 10^3 \times 500 \times 10^{-6}}{628}$$

$$Q = 12.5$$

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